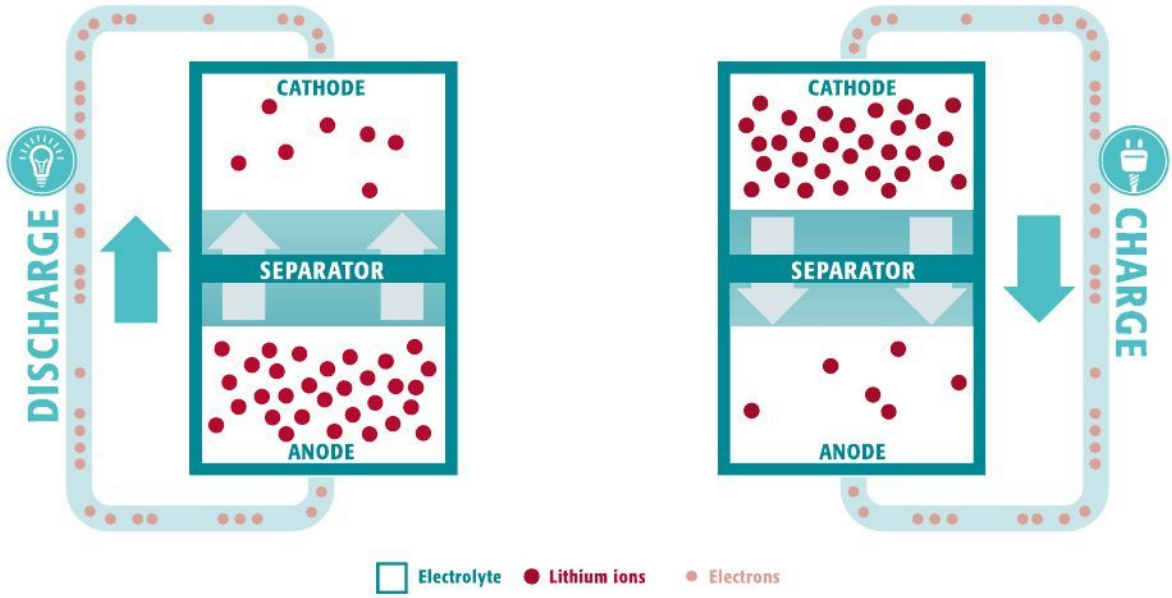
Lithium-ion batteries

Increasing popularity, underestimated risks

There are very few technology trends that have become so firmly established in recent years than the development of ever more powerful energy storage systems based on lithium (Li).
For about 30 years, lithium-ion batteries and accumulators have been conquering the market for energy storage and are used in more and more products and devices. It is no coincidence that the development of the Li-memory technology was awarded with the Nobel Prize for Chemistry in 2019.
Whether power tools, garden tools or e-bikes, our private and professional everyday life without lithium batteries is now unimaginable.
But this boom has a flip side. Because these energy storage systems that are so practical and efficient are anything but harmless. They can burst into flames and cause serious fires. Even specially shielded batteries can develop temperatures of over 1,000 °C, as experiments by the BDE have shown.

How lithium-ion batteries work

Like batteries, accumulators are electrochemical energy storage devices. They consist of two metal electrodes, a separator and a liquid electrolyte. When ions migrate from one electrode to the other, an electrical voltage is generated which can be used as an energy source. When charging, the ions are forced by an externally applied voltage to migrate back to the anode, so that the battery regenerates.



With higher energy density comes a greater risk of fire

With higher storage capacities and the increasing spread of lithium-ion batteries, the risk of fire has risen rapidly. The fact that this risk is not only of a theoretical nature has been made clear by a large number of damaging events in recent years. The recall actions were particularly spectacular with Samsung mobile devices or the worldwide flight bans on the Boeing 787 due to burning and exploding batteries.
The list of battery fires is also growing. Some examples prove the explosiveness:

- On 28th July 2020, a huge cloud of smoke rose above the site of a waste disposal site in Ellwangen. Two warehouses were on fire. The cause of the fire is suspected to be batteries located in yellow bags.
- On 1st June 2020 a lithium-ion battery triggers a fire in the building of a technology service provider in Ehningen. Although the fire can be extinguished quickly, the fire brigade is still on duty for many hours, due to highly dangerous hydrofluoric acid that escaped when the battery decomposed. The danger zone had to be cordoned off over a large area until the battery could be cooled and transported away in a special container.

These and many other cases of fire have one thing in common: the Li-batteries were stored or charged unattended without any special protective measures.

CAUSES OF BATTERY FIRES

The combination of high energy density with flammable electrolytes can hardly be avoided in the construction, so there is always a latent fire risk. If a Li-battery is not handled, stored and charged carefully and under supervision as intended by the manufacturer, uncontrolled discharges, spontaneous ignition with bursting of the battery and fire must be expected.

According to experts, undesired events and damage with batteries can generally be assigned to one of the following initial situations. These are considered particularly risky if the lithium-ion battery in combination with the high energy density of the battery

- is *mechanically damaged*, e.g. if an electric car or e-bike is involved in a traffic accident.
- is pre-damaged by *deep discharge*; in this case the cell becomes unstable, internal short circuits occur leading to over-heating.
- by external heat or energy sources *resulting in thermal overload*, e.g. when a cooling system is switched off in strong sunlight, which leads to an internal short circuit.
- is *electrically overloaded* during charging and discharging, e.g. due to a defect in the protective electronics.

The actual cause of burning, bursting and exploding lithium-ion batteries is the so-called thermal runaway. This technical term refers to a self-reinforcing chemical process which, once started, is very difficult to stop. The risk of such a thermal runaway is due to the construction of the batteries and the substances that are necessarily used.

The electrolyte liquids consist of organic solvents, i.e. they are flammable and their flammability is almost equal to that of petrol. In combination with the high energy densities this results in an extremely dangerous situation. Because as soon as the temperature in a single cell of a battery rises, all neighbouring cells also heat up in cascades. If the temperature reaches a limit range which, depending on the solvents used, starts at around 60°C and becomes extremely critical from 100 °C a chain reaction occurs which can no longer be controlled. Within fractions of a second the temperature rises up to 800 °C , the gases produced break the cover of the battery cell and the battery burns explosively.
Such a situation is unpredictable. Not only are there flames to contend with, but burning battery material is hurled through the air like fireworks. This not only endangers people, but can also start secondary fires, particularly if a fire spreads domino-like to neighbouring batteries, it spreads very quickly. Such a fire with lithium-ion batteries is feared among firefighters, as it is difficult to control. Even immediate cooling cannot stop the process, as the water only reaches the outer shell of the battery.



Please observe local legislation and regulations.

Lithium-ion batteries

WHAT DO **EXPERTS** SAY?

Assessments and recommendations by experts from fire brigades, statutory accident insurance companies or property insurers illustrate the importance of the correct handling of lithium-ion batteries.

> ***„A lithium-ion battery fire can have many causes and may occur in the complete life cycle of an energy storage system."*** *(DGUV)*

> ***„In terms of the numbers and severity of battery fires it seems like a miracle that nobody was killed so far."***
> *(BDE)*

> ***„Burning lithium-ion batteries represent a large and incalculable danger"*** *(bvfa)*

In particular, the need for safe storage of lithium-ion batteries is repeatedly stressed by all experts:

> ***„The batteries can be dangerous in a warehouse – once they catch fire, the fire is very difficult to control."***
> *(GDV after evaluation of fire tests)*

> ***„Even if relatively small quantities of the lithium-ion batteries stored catch fire, a fast fire spread is very likely. Partially explosive burning may cause the burning parts to be thrown around the area."***
> *(AGBF)*

Not without reason does the IFS recommend that lithium batteries should always be charged in a room with a smoke detector. However, a smoke detector neither prevents a fire nor spontaneous ignition and is not an effective alternative to preventive fire protection measures.

> ***„Defective or damaged batteries pose a particular fire hazard – for example, an ebike may be handed over by a customer to a service depot following a bicycle crash. It is critical that these are placed in protective containers or stored in outside buildings."***
> *(Provinzial Rheinland, Loss Prevention, Risk Advice Department*

> ***„Lithium batteries are the biggest challenge of the industry over the last 40 years. We know about waste management companies, in which they are responsible for 95% of the occuring incidents. There are injured employees and the damage to property is enormous. This cannot continue."***
> *(VOEB)*

With the growing number of damage claims and the assessments of many experts, it is becoming increasingly clear that companies and businesses that handle batteries without a comprehensive safety concept and without consistent implementation are faced with occupational safety, insurance and health risks liability in a highly risky situation.

Please observe local legislation and regulations.

LEGAL BASIS FOR HANDLING LITHIUM-ION BATTERIES IN USE

According to the Health and Safety at Work Act 1974 (HSWA), it is the duty of every employer to ensure, as far as is reasonably practicable, the health, safety and welfare at work of all its employees. This includes arrangements for ensuring, so far as is reasonably practicable, safety and absence of health risks in connection with the use, handling, storage and transport of articles and substances. The Control of Substances Hazardous to Health (COSHH), which recognises that using hazardous substances at work can put people's heath at risk, supports this. However, in accordance with regulation 5(1), COSHH only applies to substances that are hazardous to health. It is the Dangerous Substances and Explosive Atmosphere Regulations 2002 (DSEAR) which covers substances that are capable of producing effects on health because of their explosive or flammable properties.

Under both COSHH and DSEAR, the employer is obliged to reduce the accident and health risks for its employees and to evaluate them. Regulation 5 of DSEAR requires employers and the self-employed to assess risk associated with the use or presence of dangerous substances at work. DSEAR is specifically written to assist companies in protecting persons from the risk of fire, the risk of explosion and the risk of energetic energy release – all potential hazards of lithium-ion batteries.

To adhere to this, it is important that:
- *risk assessments* are carried out, which thoroughly evaluate all fire and explosion risks.
- on the basis of the risk assessments, *suitable protective measures and rules of conduct* for the employees are defined.
- *operating instructions* for handling lithium-ion batteries are prepared and displayed.
- the employees are instructed on the risks, the safety measures and the *established rules of conduct*.
- the *effectiveness of the protective measures taken is regularly reviewed* and corrected or extended as necessary.

According to DSEAR, a competent person must carry out the risk assessment. Only an expert can decide within the scope of an on-site inspection where and which measures become necessary. The DSEAR guidelines clearly state that technological measures are generally preferable over organisational or personal protection measures.
Further recommendations for the storage, handling and use of batteries in commercial and industrial premises are contained in the RC61 document, produced by the RISCAuthority in 2014. RC61 6.7 clearly advocates that storage areas should provide at least 60 minutes fire resistance between the stored batteries and any other part of the premises. Furthermore, RC61 8.12 states, "Where batteries are stored in a relatively small environment consideration may be given to installing a proprietary in-cabinet system to automatically detect and suppress a fire in the incipient stages. Where such a system is installed it should be monitored to allow the fire and rescue service to be alerted without delay".

RC61 3.3 also states that **any battery that has been damaged, dented or pierced should be taken out of service immediately, segregated from other batteries and stored safely (Section 7) while awaiting safe disposal.**
The relevant legislation for the mode of transport should be adhered to when transporting lithium batteries.
For example, by Road – ADR; by Rail – RID; by Air – ICAO and IATA; by Sea – IMDG etc.

In addition to the legislation and guidance, the many cases of fire and the assessments of experts (cf. the above quotations) provide sufficient reason for effective preventive measures in order to guarantee insurance cover. From the perspective of the insurance industry, RC61 clearly states Lithium batteries are particularly hazardous if not stored, charged and used correctly.

Please observe local legislation and regulations.

Lithium-ion batteries

GENERAL **SAFETY RULES**

Every employer must ensure that all employees who handle lithium-ion batteries at their workplaces:

- Never short-circuit batteries or protect battery terminals from short-circuiting (use pole caps or tape them off).
- Inspect batteries regularly and check for physical damage such as cracks.
- Do not use deformed, inflated, corroded, defective, damaged or hot batteries. These must be quarantined immediately and kept at a safe distance or stored separately until disposal.
- Never make your own battery packs or use batteries in any other way than specified by the manufacturer.
- Avoid hot work or open flames near batteries.

STORING LITHIUM-ION BATTERIES **SAFELY**

If Li-accumulators are provided at production sites, the quantities should be based on the daily or weekly quantities or shift requirements. All other supplies require suitable storage. If rechargeable batteries are not used for a longer period of time, e.g. removed over the winter, they can and should be stored professionally. In unsuitable storage conditions, the electrolyte may decompose and flammable gases may develop. The next time you try to recharge, an internal short circuit occurs and the cell can catch fire.

Fire and explosion risks can be significantly reduced with the following principles:

- Store dust-free and dry, protect from moisture
- Store in a temperature protected area, i.e. between 10 °C and 25 °C
- Protect from frost, but also from temperatures > 60 °C, never store in the sun or near hot surfaces, motors, etc.
- Store in fireproof separated areas
- Monitor storage areas by means of a suitable fire alarm system
- Do not store together with other fire accelerating products
- Maintain safety distances (up to 5 m from flammable materials, depending on battery type)

In workplaces with fire extinguishing systems, the extinguishing agent instructions in the safety data sheets must be observed.

SAFE CHARGING OF LITHIUM-ION BATTERIES

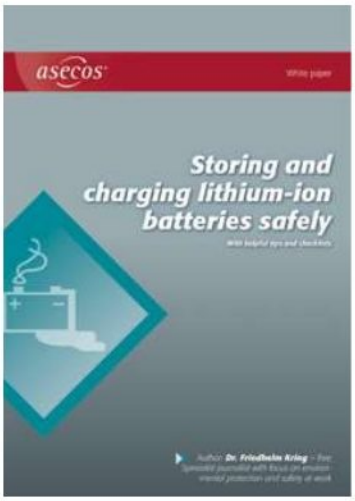
The following rules apply to safety when charging lithium-ion batteries:

- Only use with the chargers approved and intended by the manufacturer
- Only on non-combustible surfaces
- Not in the vicinity of flammable materials and fire loads
- Never cover the batteries during the charging process
- Do not charge when cold (< 0 °C)
- Never deep discharge

Although protective electronics built into the battery should prevent overcharging, such a battery management system (BMS) can fail – for example if the charger is not suitable. Thus unattended charging is always a source of danger.

Please observe local legislation and regulations.

INCREASING POPULARITY, **UNDERESTIMATED RISKS**



EXPERT KNOWLEDGE

Our new red paper is your ideal tool for obtaining quick information about lithium-ion batteries.

On 20 pages, Dr. Friedhelm Kring – freelance trade journalist with a focus on environmental protection and occupational safety – explains the essential information on lithium batteries and their hazards in an approachable manner. Tips, practical examples and checklists for the handling of lithium-ion batteries complete the red paper.

Order now!

Visit www.asecos.global to order your free copy of the red paper!

ION-LINE



FROM **RISK ASSESSMENT** TO **SAFETY CONCEPT**

Even if all the above safety rules are implemented in a company, the business owners and employers are not yet automatically legally on the safe side. The decisive factor is that not only individual measures are derived from the risk assessment, but that the business develops and implements a comprehensive safety and fire protection concept for handling lithium-ion batteries. Such an approach should answer the following 10 questions.

10 critical points of a fire protection concept for lithium-ion batteries:

- Are suitable locations and state-of-the-art technical facilities available in the workplace for the safe storage and charging of lithium batteries?
- Do these storage locations or facilities comply with the current state of the art, e.g. with regard to fire resistance, as well as the requirements for separate storage in safe distance to fire loads etc.?
- Is it ensured that the heating or burning of a battery is immediately registered (temperature sensors, smoke detectors) and indicated visually and acoustically by suitable alarm systems?
- Is the power supply to batteries in charging mode immediately and automatically interrupted in the event of increased temperatures or other safety critical situations?
- Is it guaranteed that every safety-critical condition is reliably reported to a control room, building management system or qualified personnel so that the necessary measures can be initiated immediately?
- Are both the alarm and its transmission to the intended recipients secured even outside working and operating hours, e.g. at night, at weekends or in the company holidays?
- Is it possible to remove the existing batteries quickly and safely from the building in a safety-critical situation, e.g. by using storage equipment that can be picked up by industrial trucks?
- Are the technical facilities for charging and storing batteries regularly checked for safety?
- Is the maintenance performed by qualified and competent personnel, e.g. by the manufacturer or persons certified by the manufacturer?
- Are the results of these tests documented and are the test reports kept for an appropriate period?

INSURANCE COVER AND **LIMITATION** OF **LIABILITY** ARE **AT RISK**

Temporary solutions that are not based on the current state of the art can even increase the risks if employees rely on a supposed but de facto inadequate protection against fire and explosion. Only a safety concept that ensures permanent and reliable monitoring and alarming and includes a qualified technical inspection at regular intervals, offers a sufficient degree of legal certainty.

If safety equipment such as hazardous material cabinets or fire alarm systems are not properly checked, the responsible employer and business owner not only loses insurance cover in the event of damage, but also risks the loss of the business owner liability limitation.

SAFETY CABINETS AS THE BASIS OF AN OPERATIONAL **FIRE PROTECTION CONCEPT**

Thoroughly instructed employees are undoubtedly a gain in safety. But what applies to accidents at work also applies to fire prevention: people make mistakes and individual behavioural mistakes become the cause of accidents far more often than technical failure. If you want to rule out major damage from battery fires from the start, there is no getting around technical solutions.

Please observe local legislation and regulations.

THE **ION-LINE PRO SAFETY CONCEPT**

With an integrated 3-stage warning/fire suppression system and smoke detector, the cabinets offer a high level of safety for storing and charging lithium-ion batteries.

	INCIDENTS	SYSTEM REACTIONS	SUBSEQUENT MEASURES
WARNING MESSAGE	If the interior temperature increases **above 50 °C**, the warning/fire suppression system triggers a warning message to the central control office. **Possible causes:** • Temperature build-up due to battery charging processes • Ventilation system failure	**Visual and acoustic signal output** • The warning light (red LED) is activated and permanently illuminated, the function indicator (green LED) goes out • Alarm triggers with slow tone interval **The potential-free alarm switch** • is activated, the alarm is transmitted to the building services management system	Internal qualified personnel can immediately inspect the system to take any further necessary measures. If the interior temperature decreases below 50 °C once again, the system returns to normal operations, and the visual and acoustic signals are turned off.
ALARM LEVEL 1	**Alarm level 1** is triggered when smoke begins to form in the cabinet, as soon as the smoke detector is activated. **Possible causes:** • Smoke detected without simultaneous temperature increase	**Visual and acoustic signal output** • The warning light (red LED) is activated and permanently illuminated, the function indicator (green LED) goes out • Alarm triggers with medium tone interval **The potential-free alarm switch** • is activated, the alarm is transmitted to the building services management system	Technicians (such as from the fire department) can immediately inspect the system to take any further necessary measures. If the smoke detector does not detect any further smoke production inside the cabinet, the system can be returned to normal operations by briefly unplugging it from mains voltage.
ALARM LEVEL 2	**Alarm level 2** is triggered when the smoke detector is already activated (alarm level 1) and the temperature sensor registers an interior temperature **greater than 70 °C** **Possible causes:** • Outbreak of fire	**The visual and acoustic signals change to** • warning light (red LED) switches from continuous illumination to flashing light • the alarm switches to a fast tone interval **In the BATTERY CHARGE model, at the same time** • the technical ventilation is also switched off • power to the outlet strip is turned off **The aerosol fire suppression unit** • triggers	The overall system can then only be assessed by an authorised asecos service technician and reset to normal operation if possible. At least the fire suppression unit and smoke detector must be exchanged before doing so.

EXPERT TIP: React quickly in case of a fire

Any fires which occur inside the cabinet are detected promptly, and employees can be evacuated immediately.
The warning/fire suppression system is also connected to a permanently staffed building services management, ensuring that trained rescue personnel...

- ››› can be alerted quickly and be on site in a short amount of time.
- ››› can initiate further measures immediately after completing an initial assessment of the situation.
- ››› can transport the cabinet out of the building, for instance. This prevents further major damage to the building, and protects against personal injury.

The cabinets are equipped with a transport base to ensure fast transportation. Cabinets are automatically unplugged from mains supply during transportation. Once the cabinet is outside of the building at a safe location, rescue personnel can identify any further measures necessary.
We recommend an installation at the ground level for the simplified and quick evacuation of the safety storage cabinets!

Please observe local legislation and regulations.

THE CABINETS AND **THEIR FEATURES**

When storing lithium-ion batteries, we can differentiate between passive and active storage:

ACTIVE STORAGE

In active storage, lithium-ion batteries or battery packs are charged in a cabinet with a charger or partially discharged (60–70%). Heat is generated when a lithium-ion battery charges. If this heat output is too high, a fire may occur, for instance if the lithium battery, the charger or the connection cable is defective.

Another major danger is the risk of thermal runaway of lithium-ion batteries, for instance caused by internal short circuits.

***CONCLUSION:** The risk increases when lithium-ion batteries are left unanattended to charge outside of work hours. We recommend active storage in the asecos BATTERY CHARGE safety storage cabinets.*

ION-CHARGE-90



	BATTERY CHARGE PRO	**BATTERY CHARGE**	**BATTERY CHARGE** LOCKER
Fire protection from outside (type 90)	✓	✓	✓
Fire protection from inside	✓	✓	✓
Store	✓	✓	✓
Charge	✓	✓	✓
Socket for connecting chargers	✓	✓	✓
Potential-free alarm contact for alarm forwarding	✓	✓	✓
Warning/fire suppression system	✓	✓	✓
Locker system			✓
Integrated technical ventilation	✓		
Fitted for the integration into work benches			

batteries in type 90 safety storage cabinets

PASSIVE STORAGE

In passive storage, new or used lithium-ion batteries are stored over a certain time period.

***TIP:** We recommend that new and used lithium-ion batteries are stored separately (different storage levels) in the BATTERY STORE or BATTERY STORE PRO safety storage cabinets.*

*All ION-LINE models can be found in the current brochure. Request directly at **www.asecos.global** – or simply configure your own cabinet at **www.asecos-configurator.com**.*

ION-STORE-90







BATTERY CHARGE UB	BATTERY STORE PRO	BATTERY STORE
✓	✓	✓
✓	✓	✓
✓	✓	✓
✓		
✓		
✓	✓	
✓	✓	
✓		
✓		

✓ *incl. warning and fire suppression system*

✓ *incl. warning system*

ION-LINE